

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Tuesday 21 May 2024

Afternoon (Time: 40 minutes)

Paper reference **31617H/1B**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I Biology
SECTION A: STRUCTURES AND FUNCTIONS OF CELLS AND TISSUES

You must have:
a calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
 - Section A: Structures and Functions of Cells and Tissues (Biology)
 - Section B: Periodicity and Properties of Elements (Chemistry)
 - Section C: Waves in Communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1 Figure 1 shows simple squamous epithelial cells.



Figure 1

(Source: ©PAL)

- (a) Name the piece of equipment that can be used to view the cells shown in Figure 1.

(1)

The lungs contain simple squamous epithelial cells.

- (b) Identify the area of the lungs that is lined with simple squamous epithelium.

(1)



(c) Chronic obstructive pulmonary disease (COPD) is the name for a group of lung conditions, such as emphysema, that cause breathing difficulties.

(i) Identify the main cause of COPD.

(1)

- ☐ A drinking alcohol
- ☐ B high fat diet
- ☐ C lack of exercise
- ☐ D smoking cigarettes

(ii) State what happens to the squamous epithelium in a person with emphysema.

(1)

When magnified, a squamous epithelial cell had an observed width of 4.92 mm.

(d) The squamous epithelial cell had an actual width of 82 μm .

Calculate the magnification used to observe the squamous epithelial cell.

(3)

Show your working.

magnification = X

(Total for Question 1 = 7 marks)



2 Paragraph 1 partially describes cell specialisation in animals.

(a) Complete Paragraph 1 by adding the missing words.

(2)

Cells that have the ability to develop into many different cell types are

called cells.

Cells become specialised for a particular function by a process known as

.....

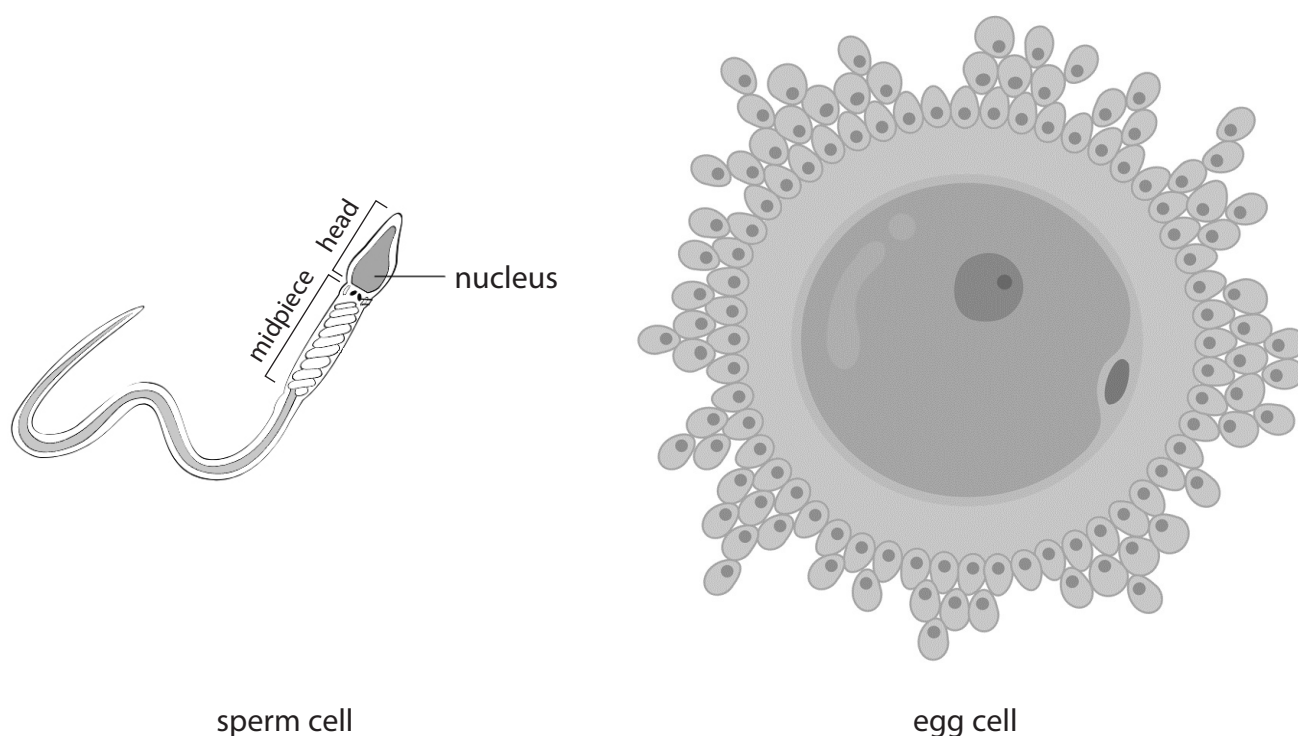
Paragraph 1



(b) Sperm and egg cells are examples of specialised animal cells.

They are specialised to enable sexual reproduction.

Figure 2 shows a diagram of a sperm cell and a diagram of an egg cell



(not to scale)

(Source: ©PAL)

Figure 2

- (i) State the importance of the cytoplasm in the egg cell for the growth of the early embryo.

(1)

- (ii) Explain why the midpiece of the sperm cell contains large numbers of mitochondria.

(2)



The egg cell is larger than the sperm cell.

The width of a sperm cell head was measured as $4\text{ }\mu\text{m}$.

The width of an egg cell was measured as 0.1 mm .

(iii) Calculate how many times wider the egg cell is than the sperm cell head.

(2)

Show your working.

The egg cell is times larger than the sperm cell head.

(Total for Question 2 = 7 marks)



3 Figure 3a shows a diagram of the cell surface of a Gram-negative bacterium.

Figure 3b shows an incomplete diagram of the cell surface of a Gram-positive bacterium.

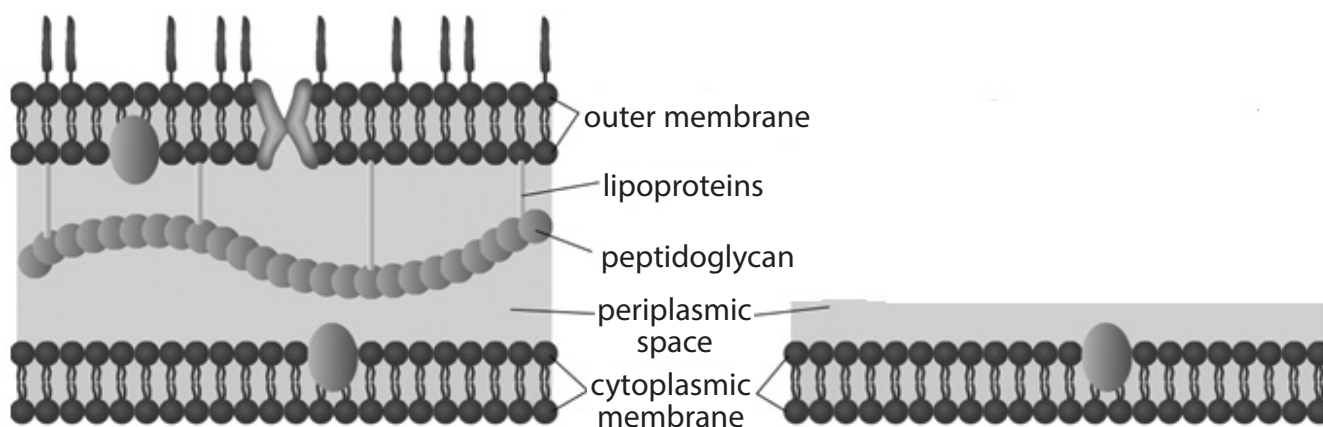


Figure 3a
cell surface of a
Gram-negative bacterium

Figure 3b
cell surface of a
Gram-positive bacterium
(incomplete)

(Source: ©PAL)

(a) Complete the diagram in Figure 3b.

(3)

The Gram staining technique can be used on a sample of bacteria.

(b) Explain why Gram staining is important.

(2)

(Total for Question 3 = 5 marks)



- 4 Paragraph 2 partially describes the action of the neurotransmitter, acetylcholine, at a synapse.

Some neurotransmitters, such as acetylcholine, can produce _____ effects and _____ effects, depending upon the type of receptors on the post-synaptic membrane.

Paragraph 2

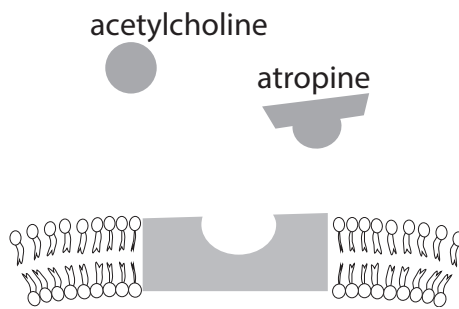
- (a) Identify the pair of words that correctly completes Paragraph 2.

(1)

- ☐ **A** active, passive
- ☐ **B** chemical, electrical
- ☐ **C** excitatory, inhibitory
- ☐ **D** good, bad



Figure 4 shows a diagram of acetylcholine and atropine at a receptor within a cell membrane.



(Source: ©PAL)

Figure 4

(4)

(Total for Question 4 = 5 marks)

5 Figure 5 shows a sarcomere in a relaxed muscle and a sarcomere in a contracted muscle.

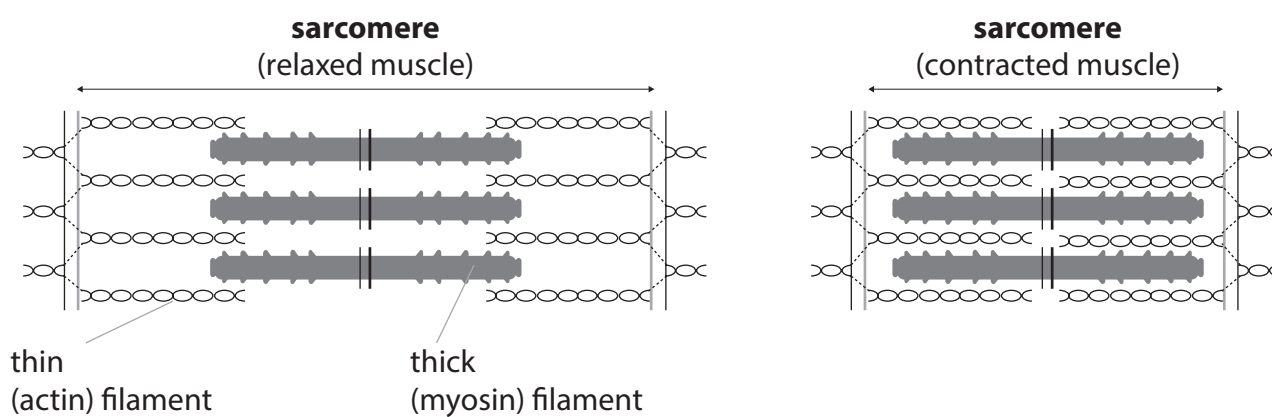


Figure 5

(Source ©PAL)

Explain the sliding filament theory of muscle contraction.

(6)

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(Total for Question 5 = 6 marks)

TOTAL FOR PAPER = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS



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